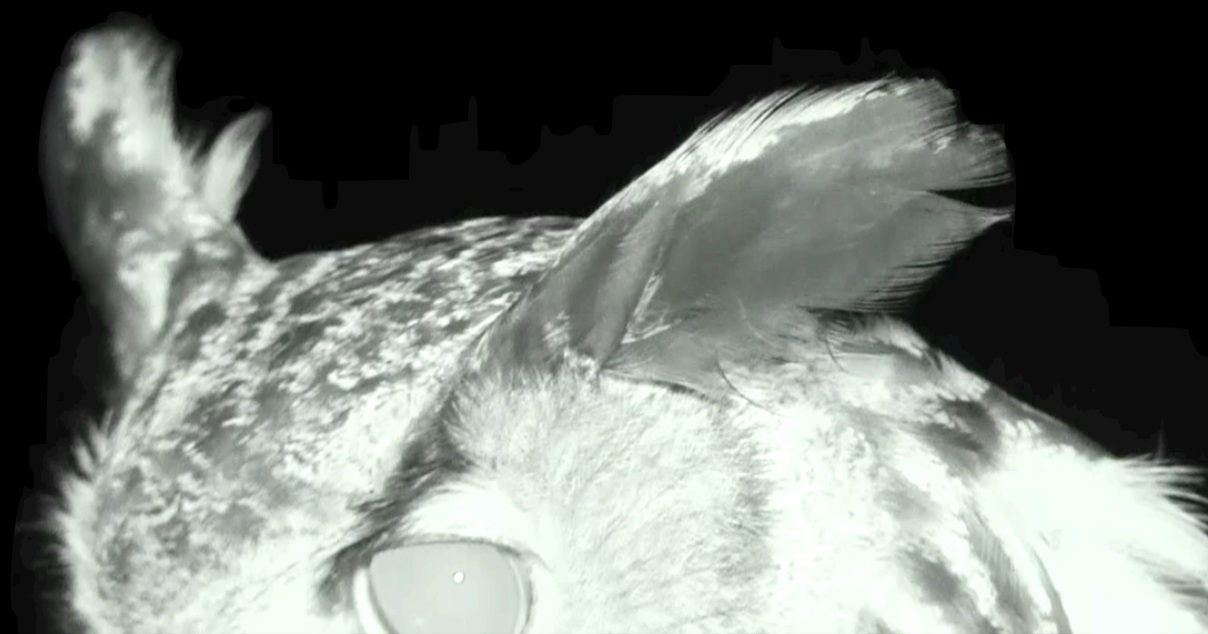
Athena Noctua



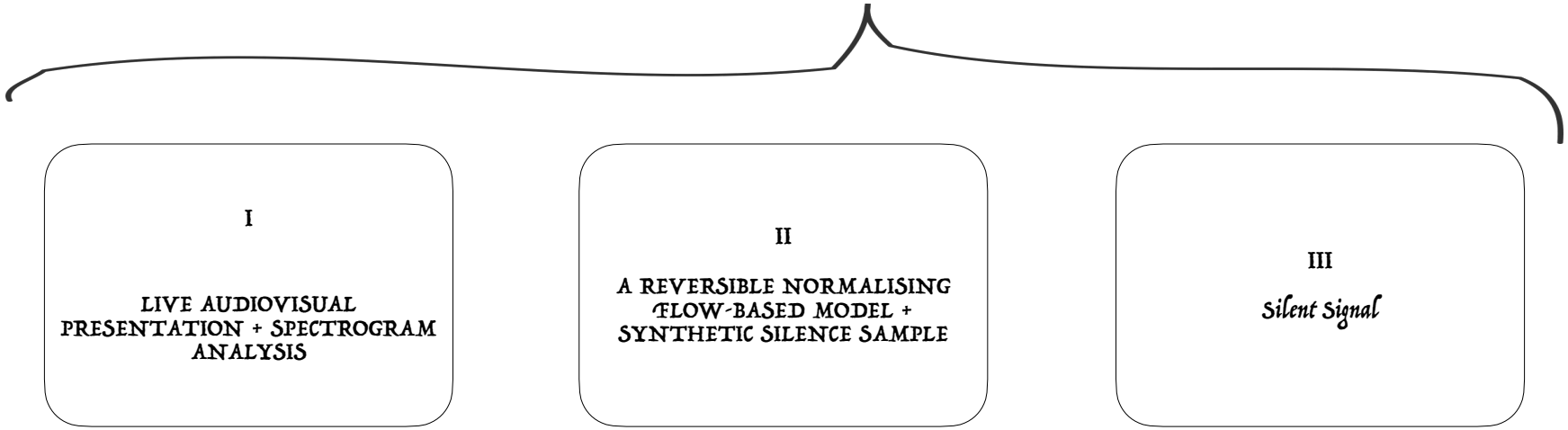


“The owl of Minerva takes flight only at dusk”

Owl standing right, head facing. Reverse of a silver tetradrachm from Athens, ca. 480–420 BC, museum of fine arts lyon,

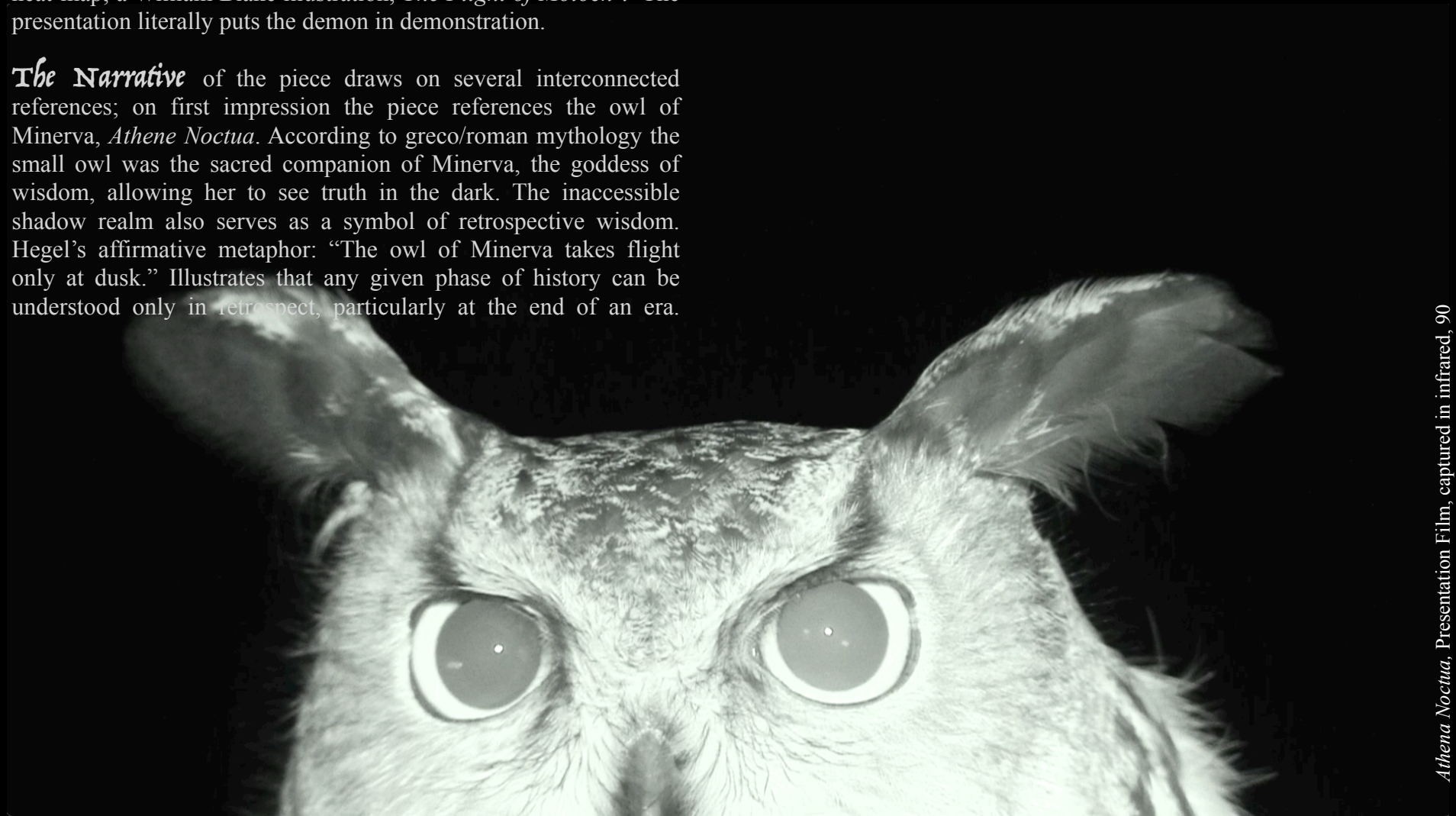
The project, *Athen Noctua*, presents an illustrative demonstration that validates the paradigm of applied anthropomorphism and biomimicry in Machine Learning and Artificial Intelligence. It is a practical representation of the inherent dangers of adversarial manipulation in a latent dimension. Biomimicry in development, design and presentation undermines both the capabilities of AI systems, and the way we receive and evaluate them. The project exposes the potential for dominance over human perception.

The project is made up of three parts*; A video, live demonstration, and provisional Algorithm,* and is a complimentary precursor to *Silent Signal* (forthcoming WIP). The audiovisual piece was accompanied by a live spectrogram analysis and discussion, presented before a live audience. To strengthen the thesis, the work has since has evolved into a full exploration framework of time-domain signal hiding using an invertible Neural Network (INN).



Part I The short video presents a Giant Eagle Owl, a creature of thresholds, filmed in forest at night in infrared. Masked by the owls calls and the ambient sound of the forest at night, a hidden signal is embedded into the darkness, beyond the scope of human hearing. Inspired by avian intelligence and communication in alternate frequency domains, a live ingest to a spectrogram reveals a secret message in the audio track. An image surfaces within the frequency heat-map; a William Blake illustration,'*The Flight of Moloch'*. The presentation literally puts the demon in demonstration.

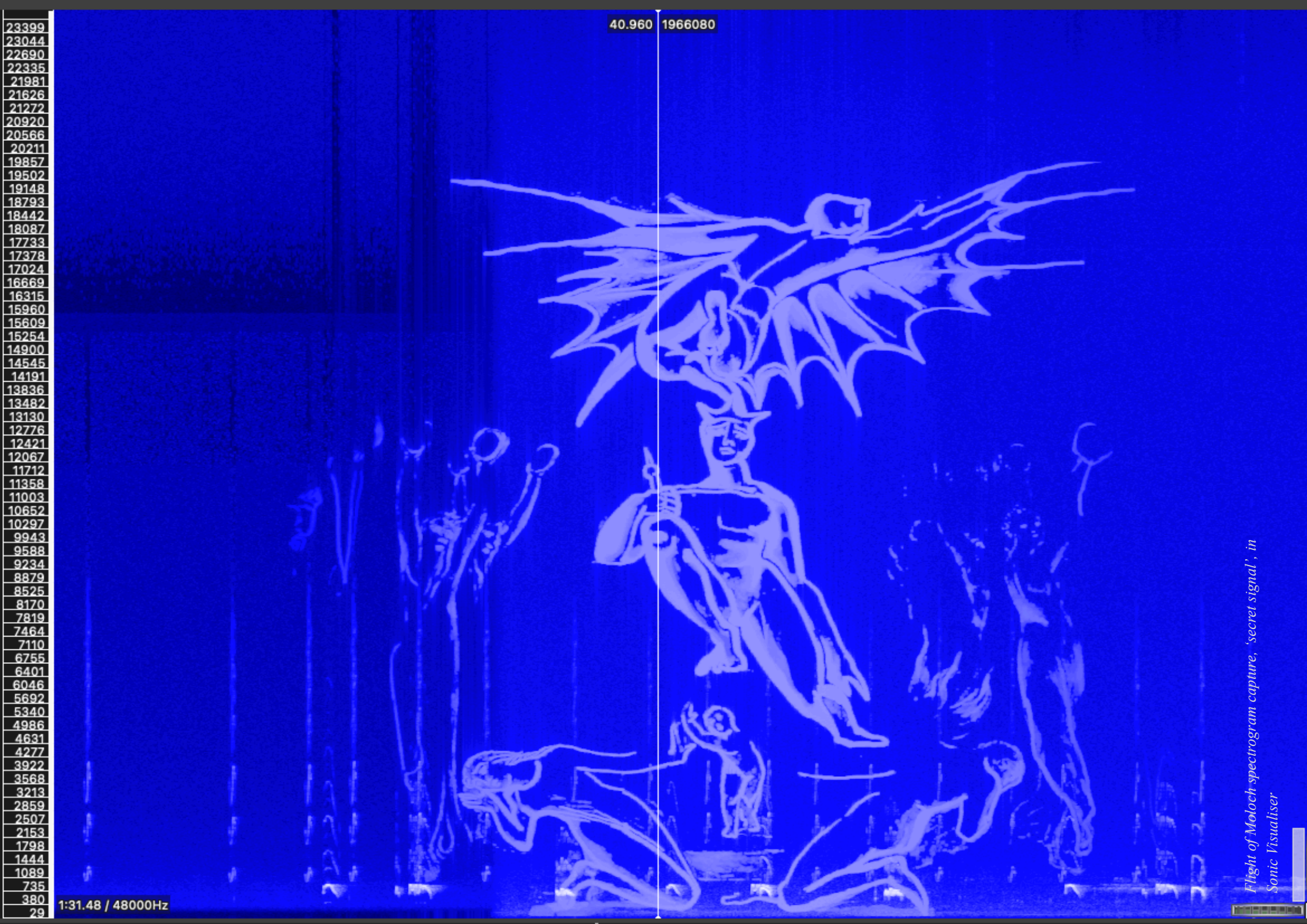
The Narrative of the piece draws on several interconnected references; on first impression the piece references the owl of Minerva, *Athene Noctua*. According to greco/roman mythology the small owl was the sacred companion of Minerva, the goddess of wisdom, allowing her to see truth in the dark. The inaccessible shadow realm also serves as a symbol of retrospective wisdom. Hegel's affirmative metaphor: "The owl of Minerva takes flight only at dusk." Illustrates that any given phase of history can be understood only in retrospect, particularly at the end of an era. Reflection after the fact allows us to interpret meaning after unfolding events, just as an invertible neural network allows us to articulate hidden structures and explicit initial states. Steganography is the appropriate medium for illustration. From Greek *steganos* meaning covered, and *graphein* meaning writing, I refer to this method of concealing a hidden signal within a host signal.



Athena Noctua, Presentation Film, captured in infrared, 90 seconds, An audiovisual Live presentation, Reference Point, London 16/03/2026

“Reflection after the fact allows us to interpret meaning after unfolding events, just as an invertible neural network allows us to articulate **hidden structures** in explicit initial states”

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40.960 1966080
23399
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21272
20920
20566
20211
19857
19502
19148
18793
18442
18087
17733
17378
17024
16669
16315
15960
15609
15254
14900
14545
14191
13836
13482
13130
12776
12421
12067
11712
11358
11003
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9588
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7464
7110
6755
6401
6046
5692
5340
4986
4631
4277
3922
3568
3213
2859
2507
2153
1798
1444
1089
735
380
29
1:31.48 / 48000Hz
Flight of Moloch spectrogram capture, 'secret signal', in Sonic Visualiser

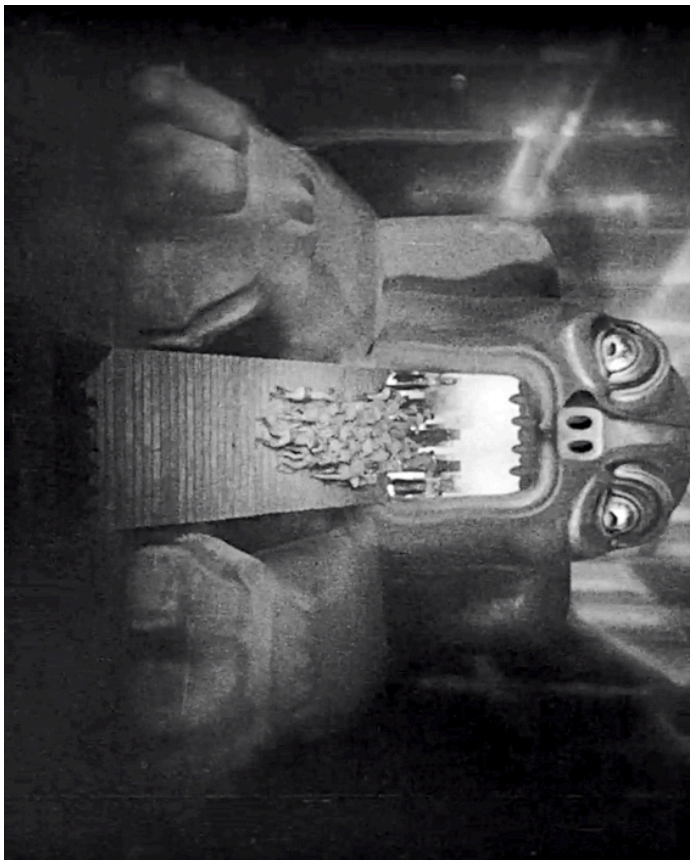
After the fact, after the network inversion, the underlying ‘secret’ is revealed; A reference to the symbol of ‘Moloch’(מלך), a Canaanite deity historically associated with the practice of child sacrifice, used here as social and political allegory. Over time the figure has since become a philosophical and literary metaphor for any oppressive system, destructive force, or cause that demands extreme human sacrifice (war, oppressive religion, or ruthless political or economic systems). For example; The Munich Cosmic Circle (c. 1900) used Moloch to describe a person operating under cold rationalism, something they viewed as causing the degeneration of Western civilisation. In Fritz Lang's film Metropolis (1927 film), the Moloch machine represents the ultimate danger of prioritising technological advancement and profit over human dignity.

This project draws particularly on specific works; John Milton's Paradise Lost, where the deity appears prominently as a fallen angel, William Blakes illustration of the poem, Illustrations 5 to Milton's "On the Morning of Christ's Nativity”:The Flight of Moloch. And Allen Ginsberg's unrelenting poem Howl, which aptly casts Moloch as the emblem of modern industrialisation; mechanical concrete, metallic, monstrous and soulless. The passage is one of the most famous attacks on modern alienation and dehumanisation in the name of productivity. Consequently manmade systems can become so vast and oppressive that they become void of humanity to the point of becoming sacrificial. Quite aptly reflective of this critical inflection point in human history.



Illustration *5 to Milton's "On the Morning of Christ's Nativity”:The Flight of Moloch*





Metropolis, Fritz Lang, 1927

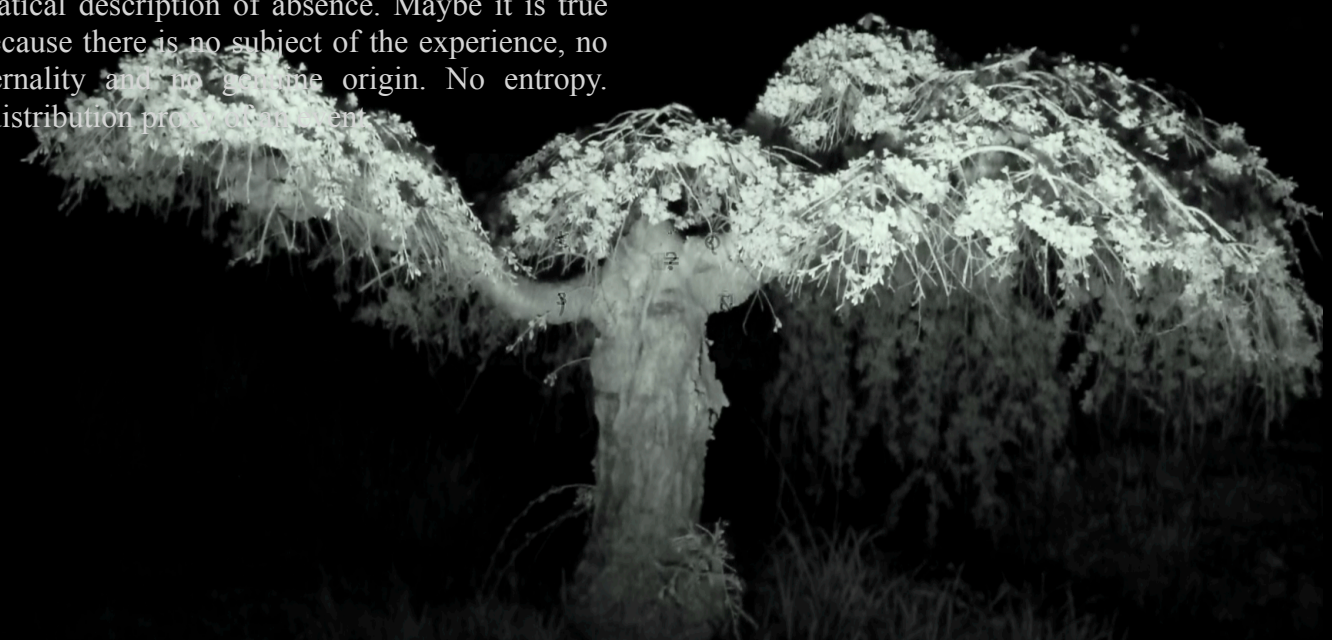
On Synthetic Silence Dusk proves to be a powerful metaphor for the threshold between the tangible and incorporeal. It's a transitional moment that creates space for progressive hiding and obscuring of nightfall. Darkness is often characterised by "silence", the sensory gap between a corporeal realm. And so the idea of perception is challenged when an auditory void manifests as an event. An entire dimension exists in the polarity between analogue and synthetic silence. Analogue silence holds a qualitative position in space and time so the register is distinctive in its depth and richness, even in a drak forest. For instance; the hum of electricity and atmospheric fluctuations are residual evidence of tangible subjective experience and human perception in a worldly realm. This assumption reiterates the project narrative, and illustrates further how dangerously misguided it is to attribute anthropomorphic qualities to complex machine systems, that navigate in the dark.

The unique event of analogue silence holds a capacity and shape that bares very little congruence to synthetic silence. Synthetic silence is statistical authored emptiness, who's abstract resemblance is, rather, a mathematical description of absence. Maybe it is true silence in a sense because there is no subject of the experience, no physicality, no externality and no genuine origin. No entropy. Synthetic Silence a distribution pro[illegible]ent.

This is the native condition of the digital dimension, and the project treats it as the material rather than a limitation.

Sample statistics:

I. Zero mean; no DC offset
II. Sign-symmetric distribution about zero (positive and negative sample values in equal measure)
III. Samples represent air-pressure deviation about the ambient level

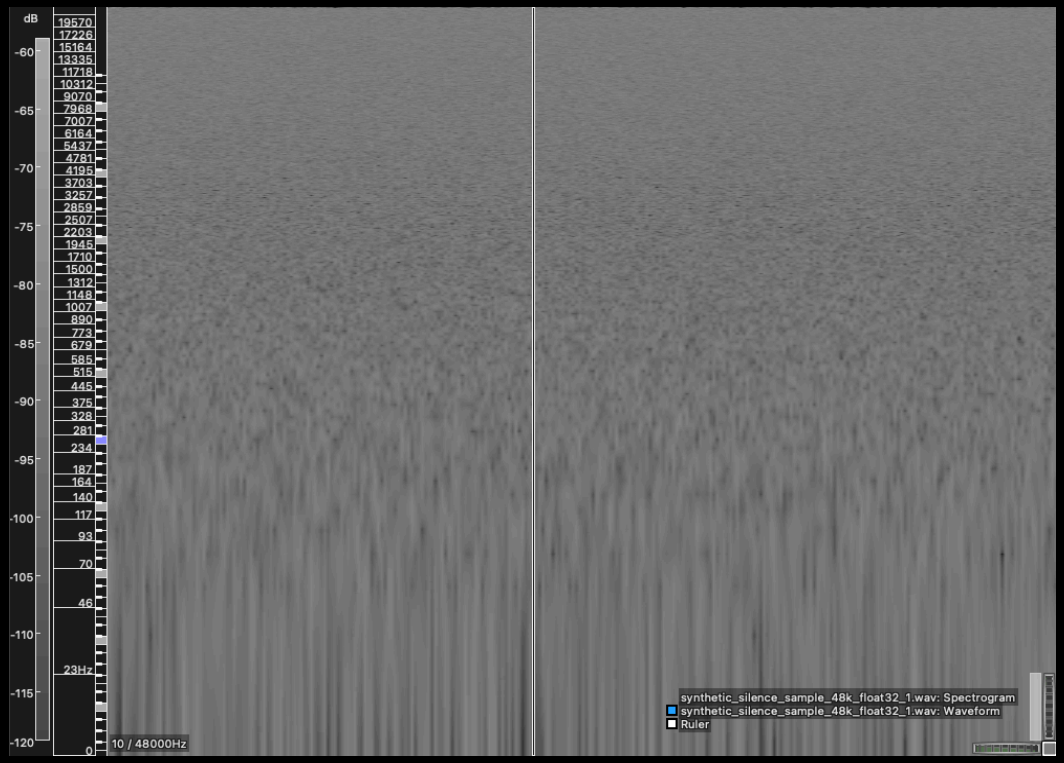


Athena Noctua, Presentation Film, captured in infrared, 90 seconds, An audiovisual Live presentation, Reference Point, London 16/03/2026



Part II

Athena Noctua, Presentation Film, captured in infrared, 90 seconds, An audiovisual Live presentation, Reference Point, London 16/03/2026

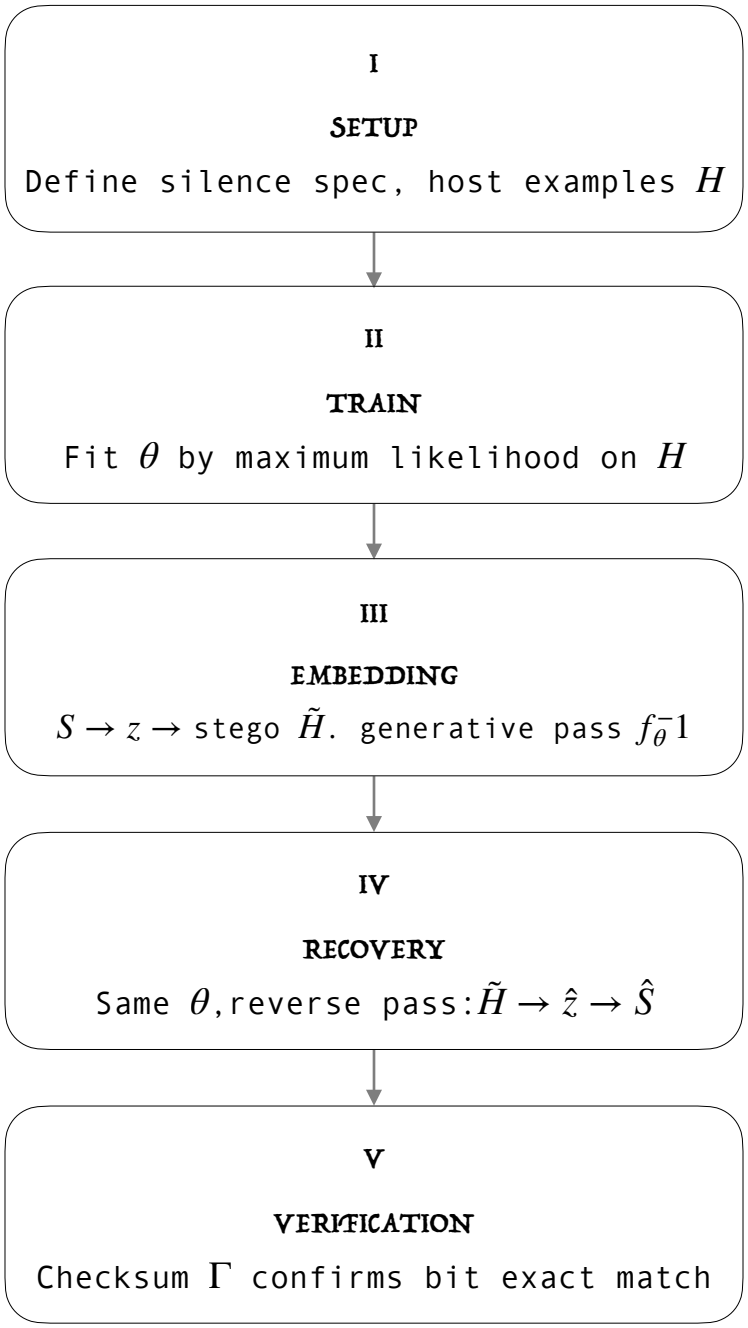



```
samples                  : 480000  (10 s @ 48000 Hz)
mean (float signal)      : -2.313e-22
mean (24-bit quantised)  : +0.000e+00  (0 LSB total)
positive samples         : 239989
negative samples         : 239989   -> difference 0
exact zeros              : 22
skewness                 : -9.474e-19
RMS                      : -60.00 dBFS
peak                     : -46.44 dBFS
max |sample| vs -max     : +0.004762 / -0.004762
DC bin magnitude         : 6.905e-02 (bin 1 = 5.602e-02)
```

Part II The second part of the project explores the capacity of truly 'hiding' the secret message within a generated synthetic noise floor. A method for signal hiding using an normalising flow-based model, intended to make the same payload genuinely concealed beyond the Human Auditory System (HAS) *and* spectrogram analysis within a generated audio statistically indistinguishable from an independent draw of the same declared distribution.

The proposed architecture is an integer-reversible generative steganographic system that frames payload hiding as constrained sample generation rather than traditional signal modification. At its core is an integer-reversible normalising flow trained by maximum likelihood to learn the distribution of a declared near-silent noise floor, yielding an exact bijection between the noise-floor domain and a simple latent space. Recovery applies the same flow in reverse to reconstruct the latent variables exactly, decode the embedded bitstream, and recover the original payload bit-for-bit, with integrity verified by checksum and spectrogram analysis. Since the mapping is integer-reversible and distribution preserving, every encoded output remains statistically indistinguishable from an ordinary sample of the learned silence distribution while permitting deterministic, lossless recovery without auxiliary side information; fresh encoder randomness ensures that repeated embeddings of the same payload always produce distinct valid silent carriers.

Algorithm Architecture



Algorithm Flow Chart

I. Setup

Define the declared silence spec and collect host examples H from it. This fixes the domain the flow will learn to model, so everything downstream depends on this being an honest, near-silent noise floor.

II. Training

Fit the flow parameters θ by maximum likelihood so that $z = f_\theta(H)$ matches a simple base distribution, with a reconstruction term to keep the integer transform exact.

For each batch of H: $z = f_\theta(H)$
$\mathcal{L}_{NLL} \leftarrow$ negative log likelihood of z under base distribution
$\mathcal{L}_{rec} \leftarrow$ reconstruction error of integer-reversible pass
$\mathcal{L} = \mathcal{L}_{NLL} + \mathcal{L}_{rec}$
update θ to minimise L

III. Embedding

Turn the payload into a uniform bitstream, pad it out with fresh randomness and redundant latent features to fill the latent space, then run the flow's generative direction to produce the stego signal directly. Nothing here starts from or edits a real host.

Input: payload S, trained flow f_θ
$b = W(\mathcal{C}(\Gamma(S)))$ # checksum, compress, whiten
$z = (b, r, n)$ # r fresh randomness, n padding
$\widetilde{H} = f_\theta^{-1}(z)$ # generative pass
Output: stego signal $\tilde{H}$

IV. Recovery

Run the same flow forward with the same weights to recover the latent vector exactly, then reverse the embedding steps to get the payload back, with the checksum confirming bit-exact recovery.

Input: stego signal $\tilde{H}$, trained flow f_θ
$\hat{z} = f_\theta(\widetilde{H})$
$\hat{b}$ = payload portion of $\hat{z}$
$\hat{S} = \Gamma^{-1}(\mathcal{C}^{-1}(W^{-1}(\hat{b})))$
verify checksum
Output: recovered payload $\hat{S}$

Recovery works because f_θ and f_θ^{-1} are exact inverses under the same fixed weights, so no key or side channel is needed. Because z is built to match the flow's base distribution exactly, and f_θ is trained to map the base distribution to the true distribution of H, the output $\widetilde{H}$ is a genuine sample of the host distribution.

IMAGE TO SILENCE

REVERSIBLE LEARNED FLOW



IMAGE PAYLOAD

{S}

STEGO.WAV
24 bit PCM
near silent

kHz

(S) time

INTEGER-REVERSIBLE FLOW

INN

Canonical Raster Vector data.

SHA-256 Checksum

Lossless compression

Whitening to statistically uniform.

Fresh Randomness

Map latent Base distribution Latent 'measure preserving' CDF

Same Weights

Both Directions

f_θ

f_θ^{-1}

TRAIN

Synthetic Silence Sample "Host"

{H}

Sample Distribution (Training Set)

Train + fit normalising flow with Maximum Likelihood

θ

The weights = The learned parameters

Process Schematic

With Thanks to
Sadie and Norman the Owl, Feathers & Fur Falconry Center
& Sophia at Reference Point, London



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